PyGame Pong Tutorial - Part 7

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Time required: 30 minutes

Pong

Time for music, sound effects, a game over menu, and a real ping pong game.

pong_assets.zip is attached to this assignment. Unzip it into a folder underneath your game folder called **assets**

1. Save **pong_6.py** as **pong_7.py**
2. Add the following code.

```
"""
    Name: pong_7.py
    Author:
    Date:
    Purpose: Add sound and game over
"""
# pip install pygame-ce
import pygame
# pip install pygame-menu
import pygame_menu as pm
# Import sys.exit to cleanly exit program
from sys import exit
from random import randint
from time import sleep
import config
from paddle import Paddle
```

```python
class Pong:

    def __init__(self):
        # pre initalize mixer with larger buffer size for better performance
        pygame.mixer.pre_init(
            44100,  # frequency (Hz)
            16,     # bit depth
            2,      # number of channels, 1 mono, 2 stereo
            4096    # buffer size, larger to optimize music playback.
        )

        # Initialize pygame
        pygame.init()
```

```python
        self.computer = Paddle(
            config.WIDTH - 15,          # x coordinate
            (config.HEIGHT - 100) // 2  # y coordinate
        )
        self.computer_speed = 3

        self.score_font = pygame.font.SysFont("freesansbold", 18)
        self.player_score = 0
        self.computer_score = 0

        # Load background music file into memory
        pygame.mixer.music.load(
            './assets/inspiring-and-uplifting-indie-rock.mp3'
        )

        # Set volume to 30%, range from 0.0 (mute) to 1.0 (full volume)
        pygame.mixer.music.set_volume(0.2)

        # Play in a loop until stopped
        pygame.mixer.music.play(-1)
```

Add the following game_over method.

```python
# ------------------------- DISPLAY GAME OVER ----------------------------#
    def game_over(self):
        """Display game over menu using the Pygame Menu library"""
        # Stop background sound
        pygame.mixer.music.stop()

        # Play crash sound
        crash = pygame.mixer.Sound('./assets/game_over.wav')
        crash.play()
        crash.set_volume(0.3)

        # Wait 2 second while crash plays
        sleep(2)

        # Define a menu object for the game over screen
        game_over = pm.Menu(
            title="Game over",      # Set title menu to "Game over"
            width=config.WIDTH,     # Set to width of game surface
            height=config.HEIGHT,   # Set to height of game surface
            # Set the theme of the menu to an orange color scheme
            theme=pm.themes.THEME_SOLARIZED
        )
```

```python
        # Display final score
        game_over.add.label(f"Player Score: {self.player_score}")
        game_over.add.label(f"Computer Score: {self.computer_score}")

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Play Again?",     # Button text
            action=main              # Call main() to start over
        )

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Exit",            # Button text
            action=pm.events.EXIT    # Exit the game when clicked
        )

        # Run the main loop of the game over menu on the specified surface
        game_over.mainloop(self.surface)
```

Check Collision

Modify the check collision method.

```python
# --------------------- CHECK COLLISION ----------------------------------#
    def check_collision(self):
        """Check for all collisions"""
        # Check for collision with left or right wall
        # Subtract ball radius to bounce off the edge of the ball
        if self.ball.left < 0 or self.ball.right >= config.WIDTH:

            # Ball goes off the table
            self.game_over()

        # Check for collision with top or bottom wall
        if self.ball.top < 0 or self.ball.bottom >= config.HEIGHT:

            # Reverse y direction multiply by -1
            self.ball_speed_y = self.ball_speed_y * -1

        # Ball collision with paddles
        if self.ball.colliderect(self.player):
            # Reverse ball direction
            self.ball_speed_x *= -1
            self.player_score += 1

            # Play ball bounce sound
            crash = pygame.mixer.Sound('./assets/hit.wav')
            crash.play()
            crash.set_volume(0.3)

        elif self.ball.colliderect(self.computer):
            # Reverse ball direction
            self.ball_speed_x *= -1
            self.computer_score += 1

            # Play ball bounce sound
            crash = pygame.mixer.Sound('./assets/hit.wav')
            crash.play()
            crash.set_volume(0.3)
```

```python
# ---------------------------- MAIN PROGRAM -------------------------
def main():
    # Create game instance/object
    pong = Pong()
    # Start the game with the run_game method
    pong.game_loop()


main()
```

Example run:

Tada, a real game!

There is always room for improvement.

1. Add more difficulty levels.
2. Keep track of the highest score between games.
3. Add more music.
4. Etc.

Assignment Submission

Zip up the program files folder and submit in Blackboard.